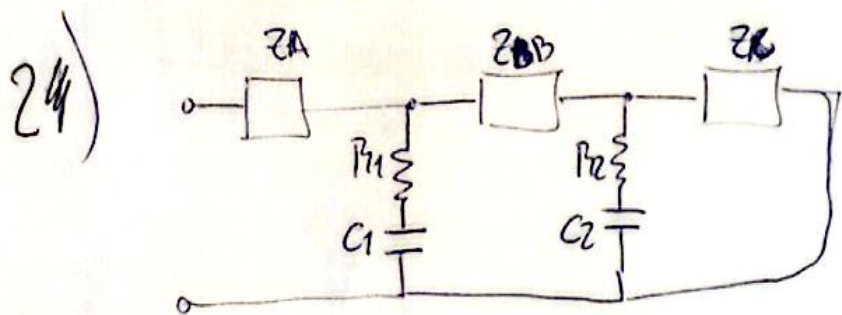


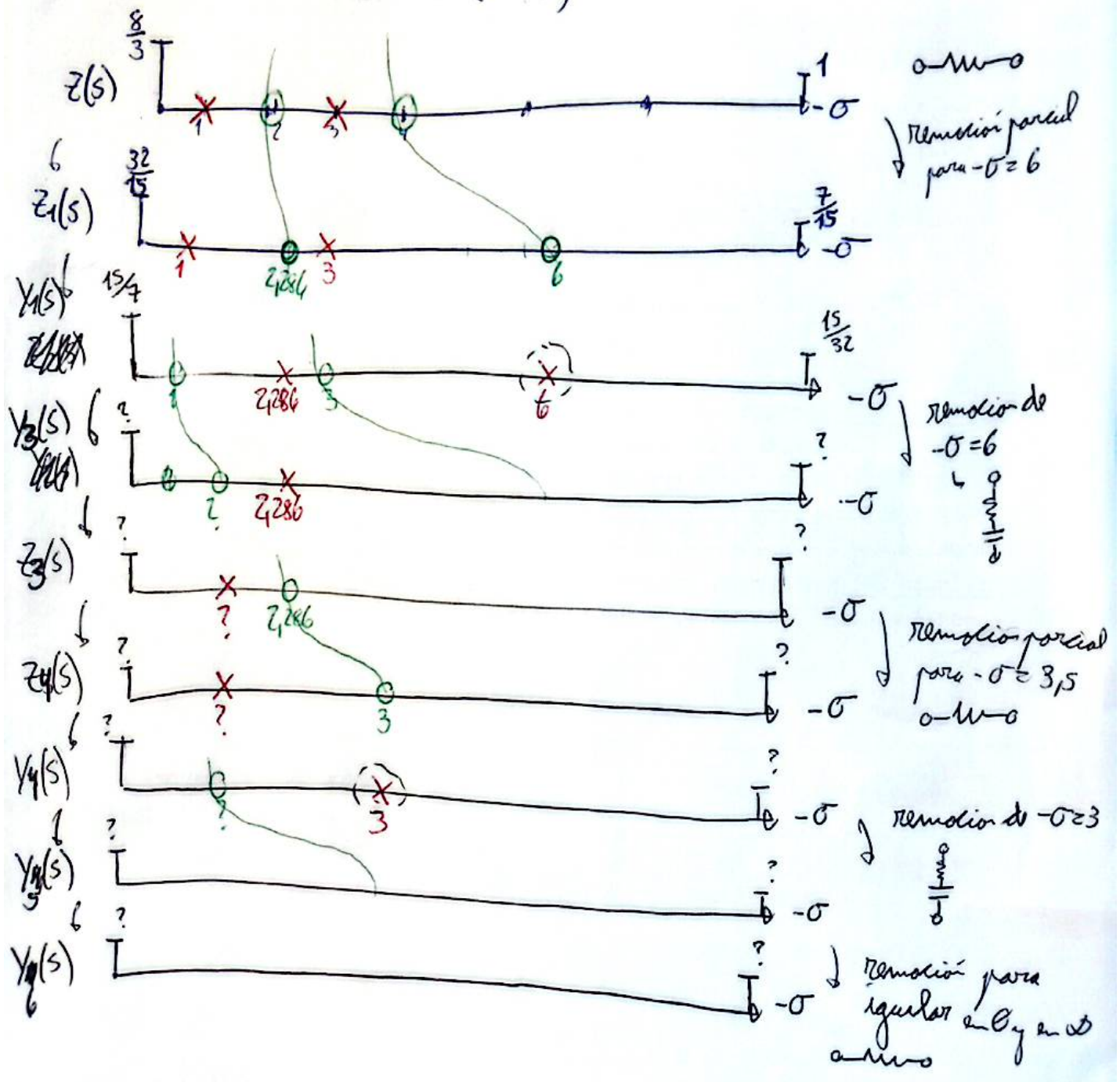
2185THC
 nuevo TP10
 (1)



$$R_1 \cdot C_1 = \frac{1}{6} \rightarrow \sigma = 6$$

$$R_2 \cdot C_2 = \frac{2}{7} \rightarrow \sigma = \frac{7}{2} = 3.5$$

$$Z(s) = \frac{s^2 + 6s + 8}{s^2 + 4s + 3} = \frac{(s+2) \cdot (s+4)}{(s+1) \cdot (s+3)}$$



Micro TPO
(2)

$$[Z_2]_{s=-6} = [Z(s) - R'_\infty]_{s=-6} = 0$$

$$R'_\infty = [Z(s)]_{s=-6} \rightarrow R'_\infty = \frac{8}{15} \rightarrow Z_A = \frac{8}{15} \rightarrow 0 \text{ mho}$$

$$Z_1(s) = Z(s) - Z_A(s)$$

$$Z_1(s) = \frac{s^2 + 6s + 8}{s^2 + 4s + 3} - \frac{8}{15} \rightarrow Z_1(s) = \frac{(s^2 + 6s + 8) - (s^2 \frac{8}{15} + 4 \frac{8}{15} s + 3 \frac{8}{15})}{s^2 + 4s + 3}$$

$$Z_1(s) = \frac{s^2 \cdot \frac{7}{15} + s \cdot \frac{58}{15} + \frac{32}{5}}{s^2 + 4s + 3} = \frac{(s+6) \cdot (s+2,286)}{(s+1) \cdot (s+3)}$$

$$Y_1(s) = \frac{(s+1) \cdot (s+3)}{(s+6) \cdot (s+2,286)}$$

me quiero sacar $\sigma = 6$

$$K_1 = \lim_{s \rightarrow -6} Y_1(s) \cdot \frac{(s+6)}{s} = \lim_{s \rightarrow -6} \frac{(s+1) \cdot (s+3)}{(s+2,286) \cdot s} \rightarrow K_1 = \frac{1250}{1857}$$

$$Y_2(s) = \frac{\frac{1250}{1857} \cdot s}{(s+6)} = \frac{1}{\frac{1857}{1250} + \frac{1}{s \cdot \frac{1250}{1857} \cdot \frac{1}{6}}} = \frac{1}{1,4856 + \frac{1}{s \cdot 0,1122}}$$

$$Y_3(s) = Y_1(s) - Y_2(s)$$

$$Y_3(s) = \frac{s^2 + 4s + 3}{(s+6) \cdot (s+2,286)} - \frac{\frac{1250}{1857} \cdot s}{(s+6)} = \frac{(s^2 + 4s + 3) - (\frac{1250}{1857} \cdot s) \cdot (s+2,286)}{(s+6) \cdot (s+2,286)}$$

$$Y_3(s) = \frac{s^2 + 4s + 3 - \frac{1250}{1857} s^2 - \frac{1905}{1238} s}{(s+6) \cdot (s+2,286)} = \frac{s^2 \cdot \frac{607}{1857} + \frac{3047}{1238} s + 3}{(s+6) \cdot (s+2,286)}$$

$$Y_3(s) = \frac{(s+6) \cdot (s+1,5296)}{(s+6) \cdot (s+2,286)} \rightarrow Y_3(s) = \frac{(s+1,5296)}{(s+2,286)}$$

$$Z_3(s) = \frac{(s + 2,286)}{(s + 1,5296)}$$

Unidad 10
(3)

Remoción parcial para $\sigma = 3,5$

$$Z_4(s) = \left[Z_3(s) - R'_{\infty} \right]_{\sigma=3,5} \quad R'_{\infty} = \left[Z_3(s) \right]_{s=3,5} \quad Z_B = 0,616$$

$$Z_4(s) = Z_3(s) - Z_B$$

$$Z_4(s) = \frac{(s + 2,286)}{(s + 1,5296)} - 0,616 = \frac{(s + 2,286) - (s \cdot 0,616 + 0,992)}{(s + 1,5296)}$$

$$Z_4(s) = \frac{s \cdot 0,384 + 1,344}{s + 1,5296} \quad Z_4(s) = \frac{0,384 \cdot (s + 3,5)}{(s + 1,5296)}$$

$$Y_4(s) = \frac{125}{48} \cdot \frac{(s + 1,5296)}{(s + 3,5)} = \frac{s \cdot 2,6 + 3,983}{(s + 3,5)}$$

$$K_1 = \lim_{s \rightarrow -3,5} Y_4(s) \cdot \frac{(s + 3,5)}{s} = \lim_{s \rightarrow -3,5} \frac{125}{48} \cdot \frac{(s + 1,5296)}{s} \quad K_1 = 1,466$$

$$Y_{22}(s) = \frac{1,466 \cdot s}{s + 3,5} = \frac{1}{\frac{1}{1,466} + \frac{1}{s \cdot 0,4188}} \rightarrow \begin{array}{c} 0 \\ \uparrow \\ 1,466 \\ \downarrow \\ 0,4188 \end{array}$$

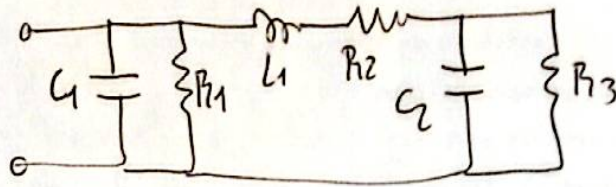
$$Y_5(s) = Y_4(s) - Y_{22}(s)$$

$$Y_5(s) = \frac{s \cdot 2,6 + 3,983}{(s + 3,5)} - \frac{1,466 \cdot s}{(s + 3,5)} = \frac{s \cdot 1,134 + 3,983}{s + 3,5} \quad Z_5(s)$$

$$Y_5(s) = \frac{1,134 \cdot (s + 3,5)}{(s + 3,5)} \rightarrow Y_5(s) = 1,134 \rightarrow Z_5(s) = 0,88$$

3)

$$Z(s) = \frac{s^2 + s + 1}{(s^2 + 2s + 5) \cdot (s + 1)}$$



$$Y(s) = \frac{1}{Z(s)} = \frac{(s^2 + 2s + 5) \cdot (s + 1)}{s^2 + s + 1}$$

Removiendo la derivación en el ∞

$$K_{\infty} = \lim_{s \rightarrow \infty} Y(s) \cdot \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{s^3 + 3s^2 + 7s + 5}{(s^2 + s + 1) \cdot s} = \lim_{s \rightarrow \infty} \frac{s^3 + 3s^2 + 7s + 5}{s^3 + s^2 + s}$$

$$K_{\infty} = 1 \rightarrow C_1 = 1 \rightarrow Y_1(s) = s \cdot 1 \rightarrow \frac{0}{s} = 1$$

$$Y_2(s) = Y(s) - Y_1(s)$$

$$Y_2(s) = \frac{s^3 + 3s^2 + 7s + 5}{s^2 + s + 1} - s \rightarrow Y_2(s) = \frac{(s^3 + 3s^2 + 7s + 5) - (s^3 + s^2 + s)}{s^2 + s + 1}$$

$$Y_2(s) = \frac{2s^2 + 6s + 5}{s^2 + s + 1}$$

Numero TP10
(5)

↓ al ser $Y(\infty)$ finito, solo una P1 en derivación

↓ $Y_3(s) = \lim_{s \rightarrow \infty} Y_2(s) \rightarrow Y_3(s) = 2 \rightarrow \left\{ \frac{1}{2} \right\}$

↓ $Y_4(s) = Y_2(s) - Y_3(s)$

↓ $Y_4(s) = \frac{2s^2 + 6s + 5}{s^2 + s + 1} - 2 = \frac{(2s^2 + 6s + 5) - (2s^2 + 2s + 2)}{s^2 + s + 1}$

↓ $Y_4(s) = \frac{4s + 3}{s^2 + s + 1}$

↓ $Z_5(s) = \frac{s^2 + s + 1}{4(s + \frac{3}{4})}$

↓ Remover en $s \rightarrow \infty \rightarrow$ un inductor serie

↓ $K_0 = \lim_{s \rightarrow \infty} Z_5(s) \cdot s = \lim_{s \rightarrow \infty} \frac{s^2 + s + 1}{4(s + \frac{3}{4})} \cdot s$

$K_0 = \lim_{s \rightarrow \infty} Z_5(s) \cdot \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{s^2 + s + 1}{4 \cdot s^2 + 3s} \rightarrow K_0 = \frac{1}{4}$ con $\frac{0}{0} \rightarrow \frac{1}{4}$

↓ $Z_6(s) = \frac{1}{4} \cdot s$

↓ $Z_7(s) = Z_5(s) - Z_6(s) = \frac{s^2 + s + 1}{s \cdot 4 + 3} - \frac{1}{4}s = \frac{(s^2 + s + 1) - (s^2 + s \cdot \frac{3}{4})}{s \cdot 4 + 3}$

↓ $Z_7(s) = \frac{s \cdot \frac{1}{4} + 1}{s \cdot 4 + 3}$

Numero TP10
(5)

↓ al ser $Y(\infty)$ finito, nos da una P1 en derivación

↓ $Y_3(s) = \lim_{s \rightarrow \infty} Y_2(s) \rightarrow Y_3(s) = 2 \rightarrow \left\{ \frac{1}{2} \right\}$

↓ $Y_4(s) = Y_2(s) - Y_3(s)$

↓ $Y_4(s) = \frac{2s^2 + 6s + 5}{s^2 + s + 1} - 2 = \frac{(2s^2 + 6s + 5) - (2s^2 + 2s + 2)}{s^2 + s + 1}$

↓ $Y_4(s) = \frac{4s + 3}{s^2 + s + 1} \rightarrow Z_5(s) = \frac{s^2 + s + 1}{4s + 3/4}$

↓ tenemos en $s \rightarrow \infty \rightarrow$ un inductor serie

↓ $K_0 = \lim_{s \rightarrow \infty} Z_5(s) \cdot s = \lim_{s \rightarrow \infty} \frac{s^2 + s + 1}{4s + 3/4} \rightarrow K_0 = \frac{1}{4}$

↓ $Z_6(s) = \frac{1}{4} \cdot s$

↓ $Z_7(s) = Z_5(s) - Z_6(s) = \frac{s^2 + s + 1}{4s + 3} - \frac{1}{4}s = \frac{(s^2 + s + 1) - (s^2 + s \frac{3}{4})}{4s + 3}$

↓ $Z_7(s) = \frac{s \frac{1}{4} + 1}{4s + 3}$

al ser RL serie $\rightarrow$ uso K_{∞} para remoción de P_7

Inverso P10
6

$$K_{\infty} = \lim_{s \rightarrow \infty} Z_7(s) = \lim_{s \rightarrow \infty} \frac{s \cdot \frac{1}{4} + 1}{s \cdot 4 + 3} \rightarrow K_{\infty} = \frac{1}{16} \rightarrow Z_8(s) = \frac{1}{16}$$

$\rightarrow \frac{1}{16}$

$$Z_9(s) = Z_7(s) - Z_8(s)$$

$$Z_9(s) = \frac{s \cdot \frac{1}{4} + 1}{s \cdot 4 + 3} - \frac{1}{16} = \frac{(s \cdot \frac{1}{4} + 1) - (\frac{1}{4} \cdot s + \frac{3}{16})}{s \cdot 4 + 3}$$

$$Z_9(s) = \frac{\frac{13}{16}}{s \cdot 4 + 3} \rightarrow Y_9(s) = \frac{s \cdot 4 + 3}{\frac{13}{16}}$$

Remuevo C en derivación en el ∞

$$K_{\infty} = \lim_{s \rightarrow \infty} Y_9(s) \cdot \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{s \cdot 4 + 3}{\frac{13}{16} \cdot s} \rightarrow K_{\infty} = \frac{64}{13}$$

$$Y_{10}(s) = \frac{64}{13} \cdot \frac{1}{s}$$

$$\frac{64}{13} \cdot \frac{1}{s}$$

$$Y_{10}(s) = \frac{64}{13} \cdot \frac{1}{s}$$

$$Y_{11}(s) = Y_9(s) - Y_{10}(s)$$

$$Y_{11}(s) = \frac{s \cdot 4 + 3}{\frac{13}{16}} - \frac{64}{13} \cdot \frac{1}{s} = \frac{(s \cdot 4 + 3) - (\frac{64}{13} \cdot \frac{1}{s})}{\frac{13}{16}}$$

$$Y_{11}(s) = \frac{48}{13}$$

$$G = \frac{48}{13}$$

$$G = \frac{48}{13} \rightarrow \frac{48}{13}$$